Module-3

Subspace Properties: Row and Column spaces - Rank and nullity - Bases for subspace - invertibility. Application on interpolation

Row and column spaces

Row space

Let
$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} = \begin{bmatrix} r_1 \\ r_2 \\ \vdots \\ r_m \end{bmatrix}$$
$$= [c_1 \quad c_2 \quad \cdots \quad c_n]$$

For the given $A_{m\times n}$ matrix the row space of A is the subspace of $\mathbb{R}^n$ spanned by the row vectors $\{r_1, \ldots r_m\}$, denoted by $R(A)$.
$$R(A) = \left\{ \sum_{i=1}^{m} \alpha_i r_i : \text{for } \alpha_i \in \mathbb{R} \right\}$$

Column space

The column space of A is the subspace of $\mathbb{R}^m$ spanned by the column vectors $\{c_1, \ldots c_n\}$ denoted by $C(A)$.
$$C(A) = \left\{ \sum_{i=1}^{n} \alpha_i c_i : \alpha_i \in \mathbb{R} \right\}$$

Null space.

The solution set of the homogeneous eqn $AX=0$ is called the null space of A, denoted by $N(A)$. Dimension of the null space is called nullity of A.
$$N(A) = \{x : Ax = 0; x \in \mathbb{R}^n\}$$

* $R(A) = C(A^T)$ and $C(A) = R(A^T)$

$Ax = x_1 c_1 + x_2 c_2 + \cdots + x_n c_n$ for any vector

$x = (x_1, x_2, \cdots, x_n) \in \mathbb{R}^n$

$C(A) = \{Ax : x \in \mathbb{R}^n\}$

$\Rightarrow$ The system $Ax = b$ has a soln iff $b \in C(A) \subseteq \mathbb{R}^m$

Thm Let U be a row (reduced) echelon form of A.
Then $R(A) = R(U)$ and $N(A) = N(U)$

Ex

$$A = \begin{bmatrix} 1 & 2 & 0 & 2 & 5 \\ -2 & -5 & 1 & -1 & -8 \\ 0 & -3 & 3 & 4 & 1 \\ 3 & 6 & 0 & -7 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 2 & 0 & 2 & 5 \\ 0 & -1 & 1 & 3 & 2 \\ 0 & 0 & 0 & -5 & -5 \\ 0 & 0 & 0 & -13 & -13 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 2 & 0 & 2 & 5 \\ 0 & -1 & 1 & 3 & 2 \\ 0 & 0 & 0 & -5 & -5 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$ $\Rightarrow$ Row ech of A $= U$

From A

$$C(A) = \left\{ x_1 \begin{pmatrix} 1 \\ -2 \\ 0 \\ 3 \end{pmatrix} + x_2 \begin{pmatrix} 2 \\ -5 \\ -3 \\ 6 \end{pmatrix} + x_3 \begin{pmatrix} 0 \\ 1 \\ 3 \\ 0 \end{pmatrix} + x_4 \begin{pmatrix} 2 \\ -1 \\ 4 \\ -7 \end{pmatrix} + x_5 \begin{pmatrix} 5 \\ -8 \\ 1 \\ 2 \end{pmatrix} \text{ s.t. } x_i \in \mathbb{R} \right\}$$

$$R(A) = \left\{ y_1 \begin{pmatrix} 1 \\ 2 \\ 0 \\ 2 \\ 5 \end{pmatrix} + y_2 \begin{pmatrix} -2 \\ -5 \\ 1 \\ -1 \\ -8 \end{pmatrix} + y_3 \begin{pmatrix} 0 \\ -3 \\ 3 \\ 4 \\ 1 \end{pmatrix} + y_4 \begin{pmatrix} 3 \\ 6 \\ 0 \\ -7 \\ 2 \end{pmatrix} : y_i \in \mathbb{R} \right\}$$

$$N(U) = \left[\left\{ \begin{pmatrix} -2s+t \\ s-t \\ s \\ -t \\ t \end{pmatrix} : s, t \in \mathbb{R} \right\} \right] = N(A)$$

How to construct bases for the fundamental spaces

For a given matrix $A_{m \times n}$

(1) To find the basis for $R(A)$

Find the ~~rke~~ row echelon or RREF of A

If the reduced matrix U obtained without changing row.

Then the basis for $R(A)$ is the set of all rows of A in which it has a pivot in U.

& If U is obtained after doing some row exchange then $R(A)$ is the set of row in U in which has a pivot in U.

(2) Basis for $C(A)$

The bases of $C(A)$ is the set of all columns of A in which it has a pivot in U.

Ex

$$A = \begin{bmatrix} 1 & 2 & 0 & 2 & 5 \\ -2 & -5 & 1 & -1 & -8 \\ 0 & -3 & 3 & 4 & 1 \\ 3 & 6 & 0 & -7 & 2 \end{bmatrix}$$

$$U = \begin{bmatrix} 1 & 2 & 0 & 2 & 5 \\ 0 & -1 & 1 & 3 & 2 \\ 0 & 0 & 0 & -5 & -5 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

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From U

$$C(U) = \left\{ x_1 \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} + x_2 \begin{pmatrix} +2 \\ -1 \\ 0 \\ 0 \end{pmatrix} + x_3 \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} + x_4 \begin{pmatrix} 2 \\ 3 \\ -5 \\ 0 \end{pmatrix} + x_5 \begin{pmatrix} 5 \\ 2 \\ -5 \\ 0 \end{pmatrix} : x_i \in R \right\}$$

$$R(U) = \left\{ y_1 \begin{pmatrix} 1 \\ 2 \\ 0 \\ 2 \\ 5 \end{pmatrix} + y_2 \begin{pmatrix} 0 \\ -1 \\ 1 \\ 3 \\ 2 \end{pmatrix} + y_3 \begin{pmatrix} 0 \\ 0 \\ 0 \\ -5 \\ -5 \end{pmatrix} : y_i \in R \right\}$$

$\Rightarrow$ $R(A) = R(U)$

But $C(A) \neq C(U)$ $\left\{ 4^{th} \text{ element is zero in } C(U) \text{ while in } C(A) \text{ non-zero} \right\}$

For finding Null space $N(A)$

Solve $AX = 0$

or $UX = 0$

$$U = \begin{bmatrix} 1 & 2 & 0 & 2 & 5 \\ 0 & -1 & 1 & 3 & 2 \\ 0 & 0 & 0 & -5 & -5 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$\because$ 3^{rd} & 5^{th} colm will not have leading one

$\therefore$ Let $x_3 = s$, $x_5 = t$

$\Rightarrow$ $x_4 = -t$,

$x_2 = s - t$

$x_1 = -2s + t$

$$\therefore X = \begin{bmatrix} -2s+t \\ s-t \\ s \\ -t \\ t \end{bmatrix} : s, t \in R$$

Theorem Let U be a RREF form of a matrix A. Then

$$R(A) = R(U) \text{ and } N(A) = N(U)$$

* If U has 'n' non-zero rows containing the leading '1', then those rows will form basis for the row space $R(A)$.

$$\Rightarrow \dim R(A) = n$$

Ex Find bases for $R(A)$, $N(A)$ and $C(A)$ of A

$$A = \begin{array}{c} \left[\begin{array}{ccccc} 1 & 2 & 0 & 2 & 5 \\ -2 & -5 & 1 & -1 & -8 \\ 0 & -3 & 3 & 4 & 1 \\ 3 & 6 & 0 & -7 & 2 \end{array}\right] \begin{array}{c} r_1 \\ r_2 \\ r_3 \\ r_4 \end{array} \\ \begin{array}{ccccc} c_1 & c_2 & c_3 & c_4 & c_5 \end{array} \end{array}$$

RREF of A $= \begin{bmatrix} 1 & 0 & 2 & 0 & 1 \\ 0 & 1 & -1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} = U$

$\Rightarrow$ First, three rows which are LI, form basis for $R(A)$.

$$\Rightarrow \dim R(A) = 3$$

[$\because$ No permutation matrix has been used (i.e. exchange of rows) these first three rows in U are corresponding to the first 3 row in A]

For $N(A)$: Solve $UX = 0$

Let $x_3 = s$, $x_5 = t$ $\Rightarrow$ $x_1 = -2s - t$, $x_2 = s - t$, $x_4 = -t$

$$\therefore \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} -2s-t \\ s-t \\ s \\ -t \\ t \end{bmatrix} = s\begin{bmatrix} -2 \\ 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t\begin{bmatrix} -1 \\ -1 \\ 0 \\ -1 \\ 1 \end{bmatrix}$$

$= \{(-2,1,1,0,0), (-1,-1,0,-1,1)\}$ is basis for $N(U) = N(A)$

$\Rightarrow \dim N(U) = 2$

* basis for $C(A)$ can be selected from Column vectector in A not in U.

$$C(A) \neq C(U)$$

$$\dim C(A) = \dim C(U)$$

Q. $A = \begin{bmatrix} 1 & -2 & 0 & 0 & 3 \\ 2 & -5 & -3 & -2 & 6 \\ 0 & 5 & 15 & 10 & 0 \\ 2 & 6 & 18 & 8 & 6 \end{bmatrix}$

Find basis for $R(A)$ and $N(A)$ of matrix A. Also find basis for $C(A)$ by findinding a basis of $R(A^T)$

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* Let A and B are two $m \times n$ matrices. Show that $AB = 0$ iff Column space of B is a subspace of the nulspace of A.

Th^m For any $m \times n$ matrix A, the row and column space of A have the same dimension

$$\dim R(A) = \dim C(A)$$ [1st fundamental thm]

* $\text{Rank}(A) = \text{Ran}(A^T)$

$$\text{Rank } A = \dim R(A) = \dim C(A)$$

$\because \dim R(A) \leq m$

$\dim C(A) \leq n$

Cor: If A is an $m \times n$ matrix then $\text{Rank } A \leq \min\{m, n\}$

Basis for $C(A)$

Let c_1, c_2, c_3, c_4, c_5 denote the column of A.

$\because$ These column span $C(A)$.

we need to discard some of the columns that can be expressed as linear combinations of other column vector

The linear dependence

$$x_1 c_1 + x_2 c_2 + x_3 c_3 + x_4 c_4 + x_5 c_5 = 0$$

$$\text{i.e. } Ax = 0$$

holds iff $x = (x_1, \ldots, x_5) \in N(A)$

By taking $x = (-2, 1, 1, 0, 0)$ $\Big\}$ basis vector

or $x = (-1, -1, 0, -1, 1)$ of $N(A)$

we obtain the nontrivial l. d. of c_i's

$$-2c_1 + c_2 + c_3 = 0$$

$$-c_1 - c_2 - c_4 + c_5 = 0$$

The column c_3 & c_5 corresponds to the free variable in $Ax = 0$ can be written as

$$c_3 = 2c_1 - c_2$$

$$c_5 = c_1 + c_2 + c_4$$

i.e. the column vectors c_3, c_5 of A are l. c. of the column vector c_1, c_2, c_4 which correspond to the basic variables in $Ax = 0$

$\Rightarrow$ $\{c_1, c_2, c_4\}$ spans the column space $C(A)$

Defn For an $m \times n$ matrix A, the rank of A is defined to be the dimension of the row space (or the column space).

Corollary For any $m \times n$ matrix A

$$\dim R(A) + \dim N(A) = \text{rank}(A) + \text{nullity of } A = n$$

$$\dim C(A) + \dim N(A^T) = \text{rank}(A) + \text{nullity of } A^T = m$$

If $\dim N(A) = 0 \Rightarrow n = \dim R(A)$

$\Rightarrow$ A has exactly n L.I. rows and n L.I. columns.

If A is a square matrix (i.e. $m = n$) then the row vectors are L.I. iff the column vectors are L.I.

Cor. Let A be an $n \times n$ sq. matrix. Then A is invertible iff rank $A = n$.

Ex

$$A = \begin{bmatrix} 1 & 2 & 0 & 2 & 1 \\ -1 & -2 & 1 & 1 & 0 \\ 1 & 2 & -3 & -7 & 2 \\ 1 & 2 & -2 & -4 & 3 \end{bmatrix}$$

RREF $\rightarrow$

$$\begin{bmatrix} 1 & 2 & 0 & 2 & 1 \\ 0 & 0 & 1 & 3 & 1 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$\because$ 1^{st} three rows in U have leading 1's which form a basis for $R(U) = R(A)$.

$\because$ 1^{st}, 3^{rd} and 5^{th} columns of U contain leading 1's. Thus the column vectors $c_1 (1, -1, 1, 1)$, $c_3 = (0, 1, -3, -2)$ and $c_5 = (1, 0, 2, 3)$ of A, not the columns in U form a basis for $C(A)$.

$\Rightarrow$ $\text{Rank}(A) = \dim R(A) = \dim C(A) = 3$

The nullity of $A = \dim N(A) = 2$, and $\dim N(A^T) = 1$.

$\dim N(A) = \dim N(U)$

$= \text{no. of free variables in } Ux = 0$

$\dim R(A) = \dim R(U)$

$=$ no. of nonzero row vectors in U

$=$ the maximal no. of LI row vectors in A

$=$ the no. of basic variables in $Ux = 0$

$=$ the maximal no. of LI column vector of A

$= \dim C(A)$

Thm Let A be an $m \times n$ matrix of rank r. Then

1. for every submatrix C of A, rank $C \leq r$ and
2. the matrix A has at least one $r \times r$ submatrix of rank r; i.e. A has an invertible submatrix of order r.

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* $\text{rank}(A) + \text{nullity}(A) = n$

Ex

$$A = \begin{bmatrix} 1 & 2 & 0 & 2 & 1 \\ 0 & 1 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$\text{Rank}(A) = 3$

$\text{rank}(A) + \text{nullity}(A) = 5 \quad \Rightarrow \text{nullity}(A) = 2$

* $\text{rank}(A) = \text{rank}(A^T)$

$A^T = 5 \times 4$ matrix

$\text{rank}(A) = \text{rank}(A^T) = 3$

$\text{rank}(A^T) + \text{nullity}(A^T) = 4 \quad \Rightarrow \text{nullity}(A^T) = 4 - 3 = 1$

while $\text{nullity}(A) = 2$

$\Rightarrow \text{nullity}(A) \neq \text{nullity}(A^T)$

Ex

$$A = \begin{bmatrix} 1 & 1 & 4 & 1 & 2 \\ 0 & 1 & 2 & 1 & 1 \\ 0 & 0 & 0 & 1 & 2 \\ 1 & -1 & 0 & 0 & 2 \\ 2 & 1 & 6 & 0 & 1 \end{bmatrix}$$

[Rank A + nullity A = n.]

RREF
$$\begin{bmatrix} 1 & 0 & 2 & 0 & 1 \\ 0 & 1 & 2 & 0 & -1 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Rank $A = 3$, nullity $A = 2$,

$\therefore$ Sol^n of $AX = 0$

$$X = \begin{bmatrix} -2s-t \\ -2s+t \\ s \\ -2t \\ t \end{bmatrix} = s\begin{bmatrix} -2 \\ -2 \\ 1 \\ 0 \end{bmatrix} + t\begin{bmatrix} -1 \\ 1 \\ 0 \\ -2 \\ 1 \end{bmatrix}$$